

```
[ '.config', '.ipynb_checkpoints', 'Churn_Modelling.csv', 'sample_data']
...
  RowNumber  CustomerId  Surname  CreditScore  Geography  Gender  Age  \
0           1    15634602  Hargrave         619     France  Female  42
1           2    15647311     Hill         608     Spain  Female  41
2           3    15619304     Onio         502     France  Female  42
3           4    15701354     Boni         699     France  Female  39
4           5    15737888  Mitchell         850     Spain  Female  43

   Tenure  Balance  NumOfProducts  HasCrCard  IsActiveMember  \
0        2     0.00              1           1                1
1        1    83807.86              1           0                1
2        8   159660.80              3           1                0
3        1      0.00              2           0                0
4        2   125510.82              1           1                1

   EstimatedSalary  Exited
0         101348.88        1
1         112542.58        0
2         113931.57        1
3          93826.63        0
4          79084.10        0
(10000, 14)
Index(['RowNumber', 'CustomerId', 'Surname', 'CreditScore', 'Geography',
      'Gender', 'Age', 'Tenure', 'Balance', 'NumOfProducts', 'HasCrCard',
      'IsActiveMember', 'EstimatedSalary', 'Exited'],
      dtype='object')
RowNumber      0
CustomerId      0
Surname         0
CreditScore     0
Geography       0
Gender         0
```

New tab

CodSoft_Task3_CustomerChurn

Ask Gemini

colab.research.google.com/drive/1okHg82qJemh5MCTc14goT_4TXcwEVAUC#scrollTo=Wd3r-VDnFPJu

☆

CodSoft_Task3_CustomerChurnPrediction.ipynb

☆

File Edit View Insert Runtime Tools Help

Q Commands

+ Code

+ Text

▶ Run all

✓

RAM

Disk

...

RowNumber

0

CustomerId

0

Surname

0

CreditScore

0

Geography

0

Gender

0

Age

0

Tenure

0

Balance

0

NumOfProducts

0

HasCrCard

0

IsActiveMember

0

EstimatedSalary

0

Exited

0

dtype: int64

Exited

0 7963

1 2037

Name: count, dtype: int64

Customer Churn Count

count

8000

7000

6000

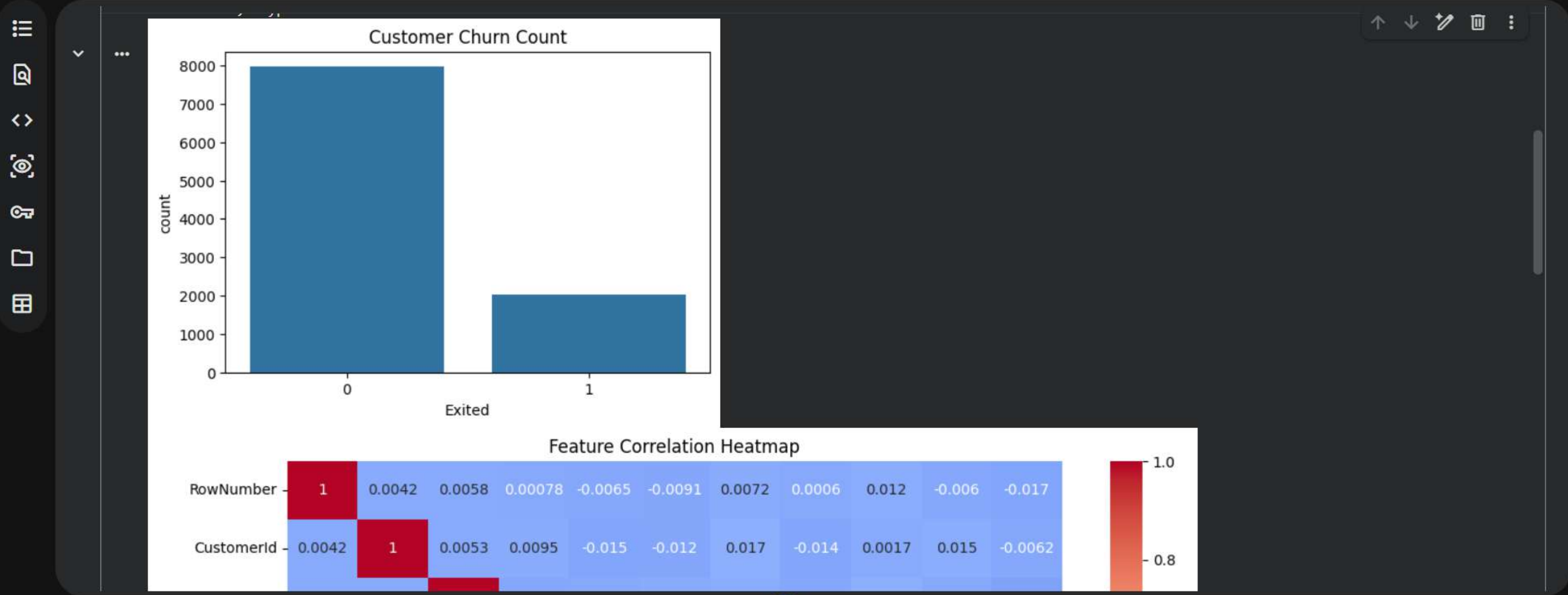
5000

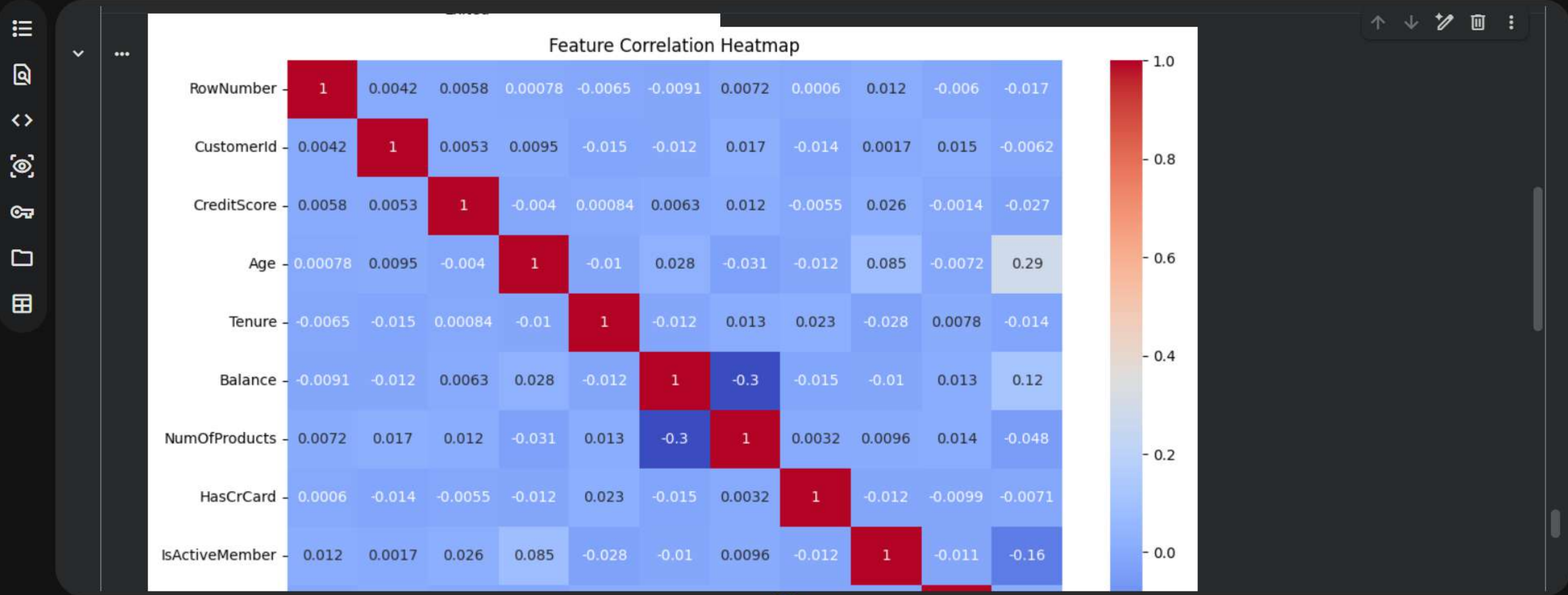
4000

Variables

Terminal

Python 3









```
4      1      1      79084.10      0
...
(8000, 10)
(2000, 10)
Training: Logistic Regression
Accuracy: 0.805
Precision: 0.58585858585859
Recall: 0.14250614250614252
F1 Score: 0.22924901185770752
      precision  recall  f1-score  support
      0      0.82    0.97    0.89    1593
      1      0.59    0.14    0.23     407

      accuracy
macro avg    0.70    0.56    0.56    2000
weighted avg 0.77    0.81    0.75    2000

Training: Decision Tree
Accuracy: 0.776
Precision: 0.45243619489559167
Recall: 0.47911547911547914
F1 Score: 0.46539379474940334
      precision  recall  f1-score  support
      0      0.86    0.85    0.86    1593
      1      0.45    0.48    0.47     407

      accuracy
macro avg    0.66    0.67    0.66    2000
weighted avg 0.78    0.78    0.78    2000

Training: Random Forest
```



```
Training: Random Forest
Accuracy: 0.86
Precision: 0.7634854771784232
Recall: 0.4520884520884521
F1 Score: 0.5679012345679012

      precision    recall  f1-score   support

      0       0.87       0.96       0.92       1593
      1       0.76       0.45       0.57        407

   accuracy          0.86          2000
  macro avg       0.82       0.71       0.74       2000
weighted avg       0.85       0.86       0.85       2000

Training: Support Vector Machine
Accuracy: 0.856
Precision: 0.8361581920903954
Recall: 0.36363636363636365
F1 Score: 0.5068493150684932

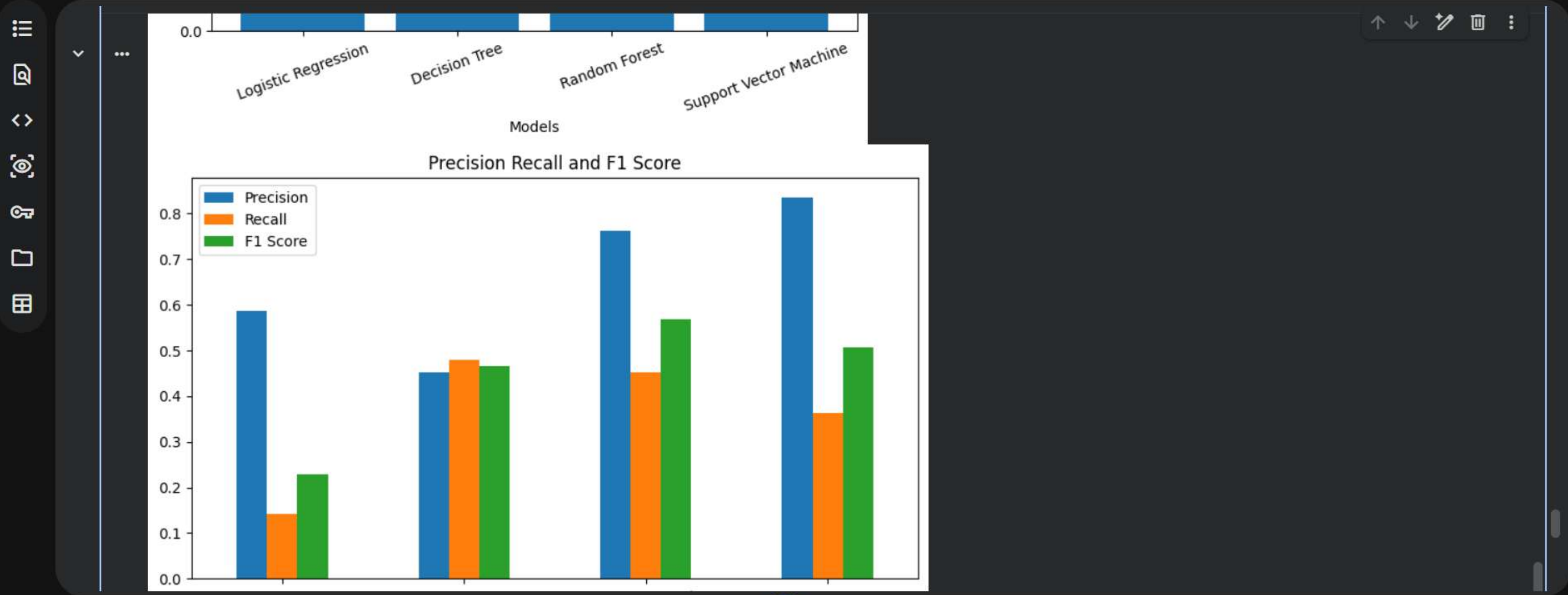
      precision    recall  f1-score   support

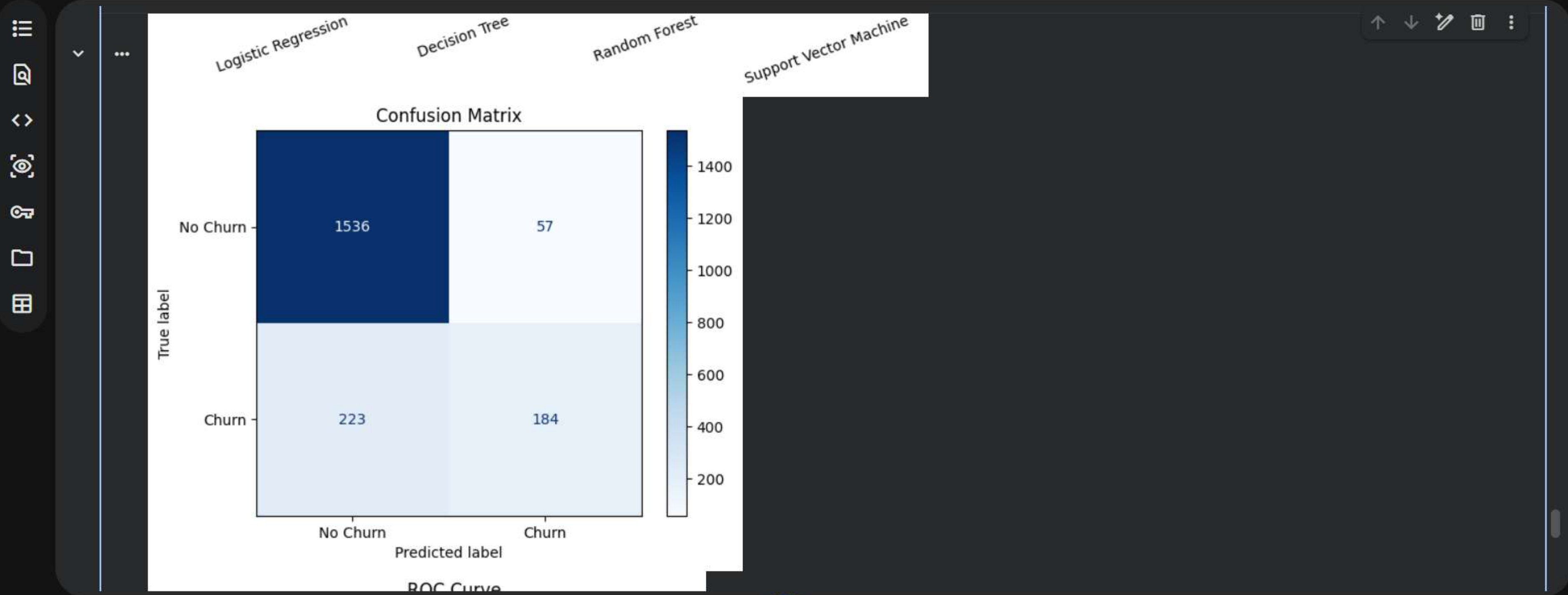
      0       0.86       0.98       0.92       1593
      1       0.84       0.36       0.51        407

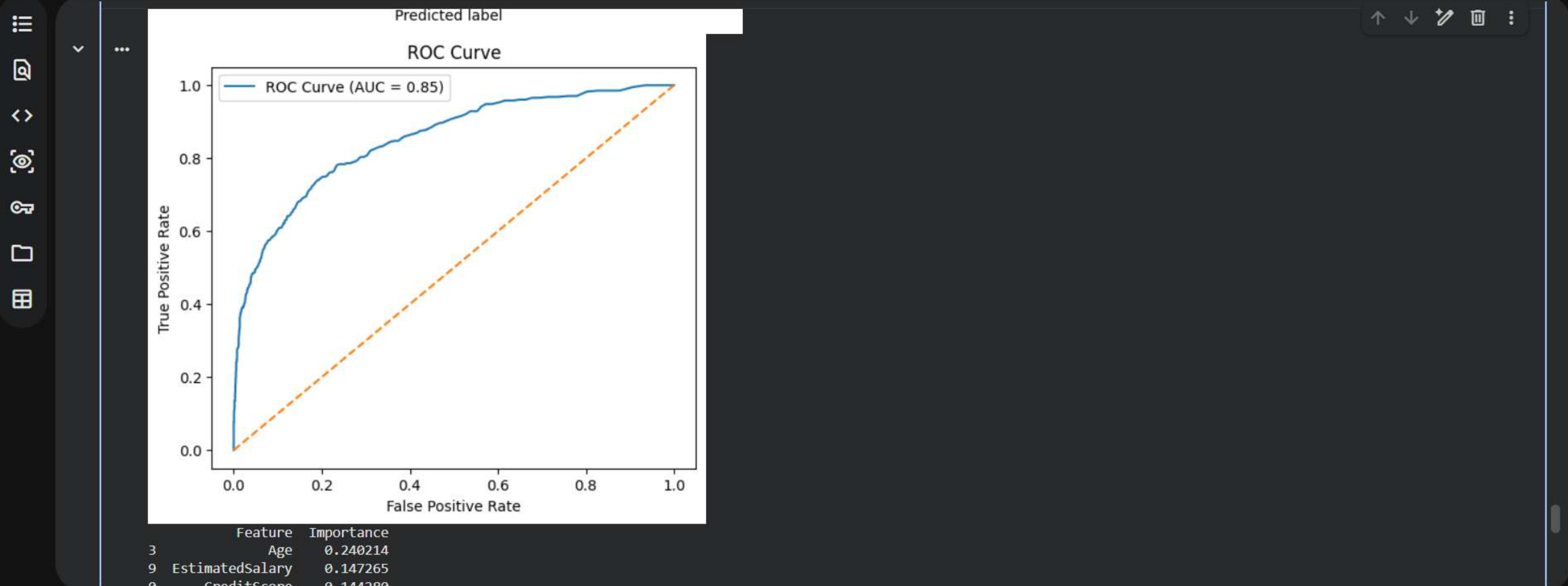
   accuracy          0.86          2000
  macro avg       0.85       0.67       0.71       2000
weighted avg       0.85       0.86       0.83       2000

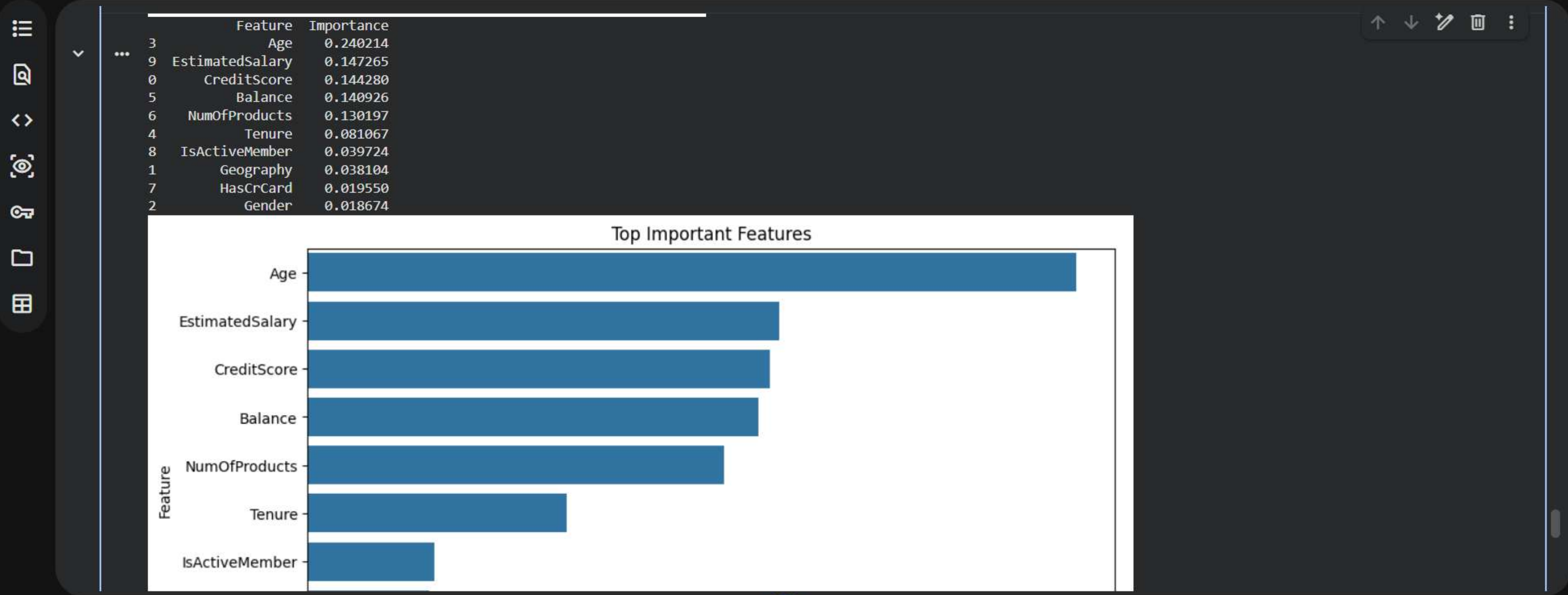
      Accuracy  Precision  Recall  F1 Score
Logistic Regression    0.805    0.585859  0.142506  0.229249
Decision Tree          0.776    0.452436  0.479115  0.465394
Random Forest          0.860    0.763485  0.452088  0.567901
```

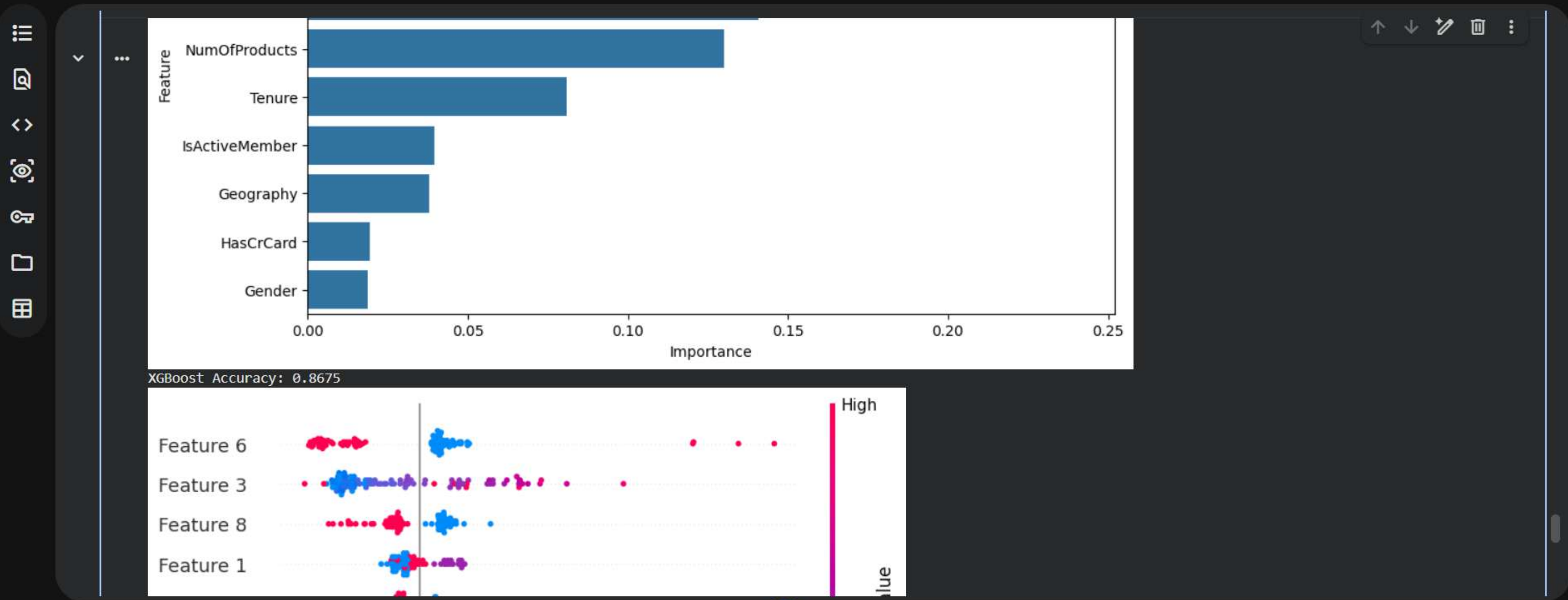


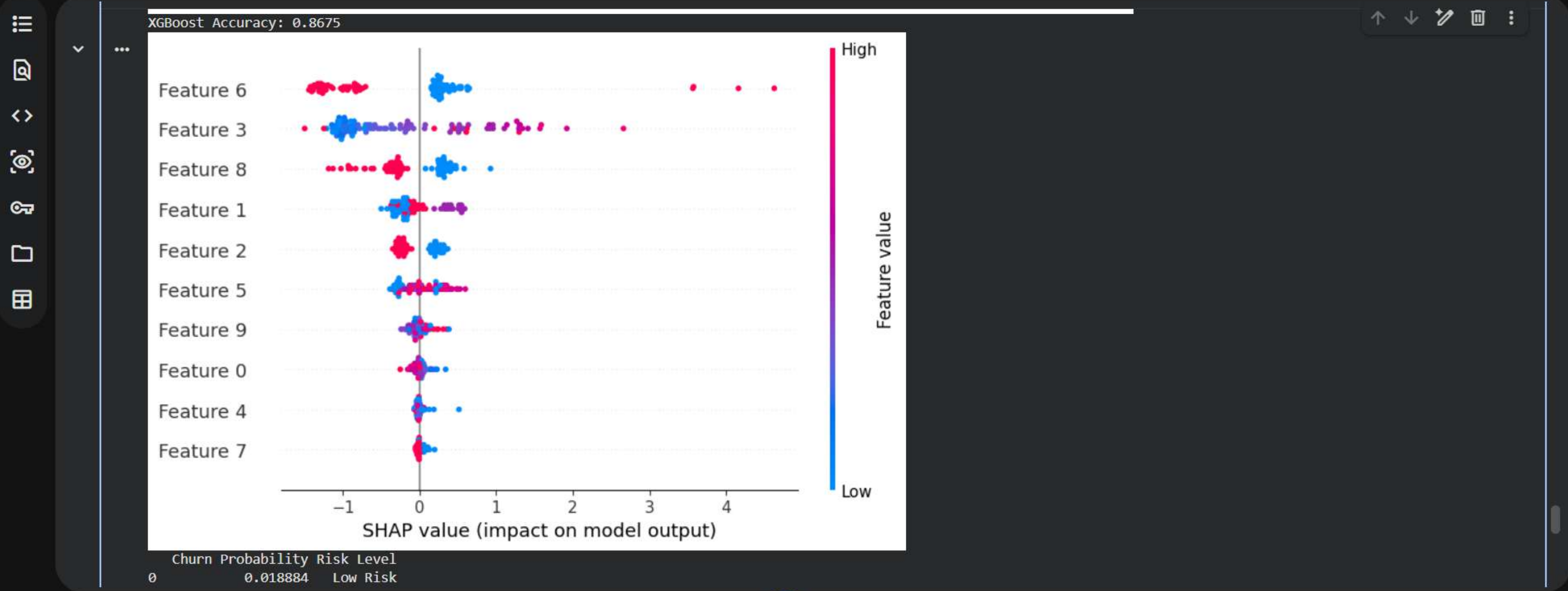













```
Churn Probability Risk Level
0 0.018884 Low Risk
1 0.077325 Low Risk
2 0.036248 Low Risk
3 0.055045 Low Risk
4 0.105673 Low Risk
----- SAMPLE CUSTOMER PREDICTIONS -----
Prediction: Customer Will Stay
Churn Probability: 0.17668892
Prediction: Customer Will Stay
Churn Probability: 0.08671696

Business Insights:

1. Customers with fewer products
   are more likely to churn.

2. Active members are less likely
   to leave the company.

3. High-risk customers can be
   targeted with retention offers.

4. The prediction system helps
   companies reduce customer loss.

Advanced Customer Churn Prediction Completed
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
```